Module 2

# Accounting Statements and Finance

FINE 3010, Financial Management

Prof. Renping Li, Tulane University

# Why Do We Need Accounting?

1. What do firms own?
   - Real assets (tangible, intangible) and financial assets (cash, securities)

# Why Do We Need Accounting?

1. What do firms own?
   - Real assets (tangible, intangible) and financial assets (cash, securities)
2. What do firms owe?

# Why Do We Need Accounting?

1. What do firms own?
   - Real assets (tangible, intangible) and financial assets (cash, securities)
2. What do firms owe?
3. How much money did firms make?
   - Income? Cash flow?

# Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?




CNBC, August 18, 2026 (3:50), "Anthropic tells investors annualized revenue run rate climbed to $65 billion in July"

# Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- We will have a clear understanding after this module




CNBC, August 18, 2026 (3:50), "Anthropic tells investors annualized revenue run rate climbed to $65 billion in July"

# Three Financial Statements

- **Balance sheet**
  - What do firms own and owe at a point in time



# Three Financial Statements

- **Balance sheet**
  - What do firms own and owe at a point in time
- **Income statement**
  - How much did firms earn in the past year



# Three Financial Statements

- **Balance sheet**
  - What do firms own and owe at a point in time
- **Income statement**
  - How much did firms earn in the past year
- **Statement of cash flows**
  - Cash inflows and outflows during the past year



# Securities and Exchange Commission (SEC)

- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure




sec.gov/edgar/search

# Securities and Exchange Commission (SEC)

- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure
- **EDGAR**, the SEC's free public database of company filings




sec.gov/edgar/search

# Securities and Exchange Commission (SEC)

- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure
- **EDGAR**, the SEC's free public database of company filings
- **10-K**, a firm's audited annual report filed with the SEC




sec.gov/edgar/search

# Goals

- Interpret the balance sheet, income statement, and statement of cash flows

# Goals

- Interpret the balance sheet, income statement, and statement of cash flows
- Distinguish between market and book values

# Goals

- Interpret the balance sheet, income statement, and statement of cash flows
- Distinguish between market and book values
- Explain why income differs from cash flow

# Goals

- Interpret the balance sheet, income statement, and statement of cash flows
- Distinguish between market and book values
- Explain why income differs from cash flow
- Understand the taxation of corporate and personal income

## Module 2, Part 1

# The Balance Sheet

# Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time

# Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time
- Components
  - **Assets**, what the firm owns and uses to generate return

# Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time
- Components
  - **Assets**, what the firm owns and uses to generate return
  - **Liabilities**, what the firm owes to others

# Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time
- Components
  - **Assets**, what the firm owns and uses to generate return
  - **Liabilities**, what the firm owes to others
  - **Shareholders' equity**, a residual term that equals assets minus liabilities

# Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time
- Components
  - **Assets**, what the firm owns and uses to generate return
  - **Liabilities**, what the firm owes to others
  - **Shareholders' equity**, a residual term that equals assets minus liabilities

$$\textbf{Assets} = \textbf{Liabilities} + \textbf{Equity}$$

# Balance Sheet

***Where does the balance sheet come from?***

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)

# Balance Sheet

***Where does the balance sheet come from?***

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)
- The balance sheet "grows" in this process

# Balance Sheet

***Where does the balance sheet come from?***

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)
- The balance sheet "grows" in this process
- The numbers are the cost of acquiring real assets, or the amount raised from selling financial assets

# Balance Sheet

***Where does the balance sheet come from?***

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)
- The balance sheet "grows" in this process
- The numbers are the cost of acquiring real assets, or the amount raised from selling financial assets
- The numbers stay at the transaction amount, even when the asset is worth more or less later

# Example, Issuing Stock to Build a Data Center ($ Billions)

| 1. Today | | | |
|---|---|---|---|
| Cash | 1 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 1 |
| **Total assets** | **1** | **Total liabilities and equity** | **1** |

# Example, Issuing Stock to Build a Data Center ($ Billions)

| 1. Today | | | |
|---|---|---|---|
| Cash | 1 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 1 |
| **Total assets** | **1** | **Total liabilities and equity** | **1** |

| 2. Issue stock | | | |
|---|---|---|---|
| Cash | 2 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

# Example, Issuing Stock to Build a Data Center ($ Billions)

**1. Today**

| | | | |
|---|---|---|---|
| Cash | 1 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 1 |
| **Total assets** | **1** | **Total liabilities and equity** | **1** |

**2. Issue stock**

| | | | |
|---|---|---|---|
| Cash | 2 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

**3. Build data center**

| | | | |
|---|---|---|---|
| Cash | 0 | Liabilities | 0 |
| Fixed assets | 2 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

# Example, Issuing Stock to Build a Data Center ($ Billions)

**1. Today**

| | | | |
|---|---|---|---|
| Cash | 1 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 1 |
| **Total assets** | **1** | **Total liabilities and equity** | **1** |

**2. Issue stock**

| | | | |
|---|---|---|---|
| Cash | 2 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

**3. Build data center**

| | | | |
|---|---|---|---|
| Cash | 0 | Liabilities | 0 |
| Fixed assets | 2 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

- *Should you revise fixed assets if the data center is now worth more than $2 billion?*

# Example, Issuing Stock to Build a Data Center ($ Billions)

**1. Today**

| | | | |
|---|---|---|---|
| Cash | 1 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 1 |
| **Total assets** | **1** | **Total liabilities and equity** | **1** |

**2. Issue stock**

| | | | |
|---|---|---|---|
| Cash | 2 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

**3. Build data center**

| | | | |
|---|---|---|---|
| Cash | 0 | Liabilities | 0 |
| Fixed assets | 2 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

- *Should you revise fixed assets if the data center is now worth more than $2 billion?* **No**

# Example, Issuing Stock to Build a Data Center ($ Billions)

**1. Today**

| | | | |
|---|---|---|---|
| Cash | 1 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 1 |
| **Total assets** | **1** | **Total liabilities and equity** | **1** |

**2. Issue stock**

| | | | |
|---|---|---|---|
| Cash | 2 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

**3. Build data center**

| | | | |
|---|---|---|---|
| Cash | 0 | Liabilities | 0 |
| Fixed assets | 2 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

- *Should you revise fixed assets if the data center is now worth more than $2 billion?* **No**
- *Should you revise shareholders' equity if someone is willing to buy the business at $4 billion?*

# Example, Issuing Stock to Build a Data Center ($ Billions)

| 1. Today | | | |
|---|---|---|---|
| Cash | 1 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 1 |
| **Total assets** | **1** | **Total liabilities and equity** | **1** |

| 2. Issue stock | | | |
|---|---|---|---|
| Cash | 2 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

| 3. Build data center | | | |
|---|---|---|---|
| Cash | 0 | Liabilities | 0 |
| Fixed assets | 2 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

- *Should you revise fixed assets if the data center is now worth more than $2 billion?* **No**
- *Should you revise shareholders' equity if someone is willing to buy the business at $4 billion?* **No**

# Balance Sheet

***Where does the balance sheet come from?***

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)
- The balance sheet "grows" in this process
- The numbers are the cost of acquiring real assets, or the amount raised from selling financial assets
- The numbers stay at the transaction amount, even when the asset is worth more or less later

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Current assets**
Assets that can generate return within one year

| Assets | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Current assets**

Assets that can generate return within one year

1. Cash and marketable securities

| Assets | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Current assets**

Assets that can generate return within one year

1. Cash and marketable securities
2. Accounts receivable
   - Amount owed by customers

| Assets | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Current assets**

Assets that can generate return within one year

1. Cash and marketable securities
2. Accounts receivable
   - Amount owed by customers
3. Inventories
   - The amount in the table is always the acquisition cost

| Assets | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Current assets**

Assets that can generate return within one year

1. Cash and marketable securities
2. Accounts receivable
   - Amount owed by customers
3. Inventories
   - The amount in the table is always the acquisition cost

| Assets | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Fixed assets**

Assets that can generate return for longer than one year

| Assets | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Fixed assets**

Assets that can generate return for longer than one year

1. Tangible fixed assets
   - Property, plant, equipment ("PP&E"), the amount in the table is always the acquisition cost
   - Accumulated depreciation, reduction in the value of assets over time

| Assets | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Fixed assets**

Assets that can generate return for longer than one year

1. Tangible fixed assets
   - Property, plant, equipment ("PP&E"), the amount in the table is always the acquisition cost
   - Accumulated depreciation, reduction in the value of assets over time

| Assets | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Fixed assets**

Assets that can generate return for longer than one year

1. Tangible fixed assets
   - Property, plant, equipment ("PP&E"), the amount in the table is always the acquisition cost
   - Accumulated depreciation, reduction in the value of assets over time
2. Intangible assets
   - Difficult to value them accurately

| **Assets** | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Fixed assets**

Assets that can generate return for longer than one year

1. Tangible fixed assets
   - Property, plant, equipment ("PP&E"), the amount in the table is always the acquisition cost
   - Accumulated depreciation, reduction in the value of assets over time
2. Intangible assets
   - Difficult to value them accurately

| **Assets** | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Fixed assets**

Assets that can generate return for longer than one year

1. Tangible fixed assets
   - Property, plant, equipment ("PP&E"), the amount in the table is always the acquisition cost
   - Accumulated depreciation, reduction in the value of assets over time
2. Intangible assets
   - Difficult to value them accurately

| Assets | |
|---|---|
| Current assets | |
| Cash and marketable securities | 2,600 |
| Accounts receivable | 500 |
| Inventories | 9,000 |
| Total current assets | 12,100 |
| Fixed assets | |
| Tangible fixed assets | |
| Property, plant, equipment | 48,200 |
| Less accumulated depreciation | 19,700 |
| Net tangible fixed assets | 28,500 |
| Intangible asset | 700 |
| Total fixed assets | 29,200 |
| **Total assets** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Current liabilities**

Obligations due within a year

| Liabilities and equity | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Current liabilities**

Obligations due within a year

1. Debt due for repayment
   - Debt due for repayment within a year

| Liabilities and equity | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Current liabilities**

Obligations due within a year

1. Debt due for repayment
   - Debt due for repayment within a year
2. Accounts payable
   - Amount owed to suppliers

| Liabilities and equity | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Current liabilities**

Obligations due within a year

1. Debt due for repayment
   - Debt due for repayment within a year
2. Accounts payable
   - Amount owed to suppliers

| Liabilities and equity | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Long-term liabilities**

Obligations due after a year

| Liabilities and equity | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Long-term liabilities**

Obligations due after a year

1. Long-term debt
   - Bonds, debt from public bond markets
   - Bank loans, debt from banks

| Liabilities and equity | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Long-term liabilities**

Obligations due after a year

1. Long-term debt
   - Bonds, debt from public bond markets
   - Bank loans, debt from banks

| Liabilities and equity | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Shareholders' equity**

A *residual* term that equals assets minus liabilities

| **Liabilities and equity** | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Shareholders' equity**

A *residual* term that equals assets minus liabilities

1. Paid-in capital
   - Amount raised from previous sales of ownership

| **Liabilities and equity** | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Shareholders' equity**

A *residual* term that equals assets minus liabilities

1. Paid-in capital
   - Amount raised from previous sales of ownership
2. Retained earnings
   - Accumulated profits kept in the business and reinvested on the shareholders' behalf

| **Liabilities and equity** | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Shareholders' equity**

A *residual* term that equals assets minus liabilities

1. Paid-in capital
   - Amount raised from previous sales of ownership
2. Retained earnings
   - Accumulated profits kept in the business and reinvested on the shareholders' behalf

| **Liabilities and equity** | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

**Shareholders' equity**

A *residual* term that equals assets minus liabilities

1. Paid-in capital
   - Amount raised from previous sales of ownership
2. Retained earnings
   - Accumulated profits kept in the business and reinvested on the shareholders' behalf

| Liabilities and equity | |
|---|---|
| Current liabilities | |
| Debt due for repayment | 200 |
| Accounts payable | 9,900 |
| Total current liabilities | 10,100 |
| Long-term liabilities | 11,300 |
| Total liabilities | 21,400 |
| | |
| Shareholders' equity | |
| Paid-in capital | 6,300 |
| Retained earnings | 13,600 |
| Total shareholders' equity | 19,900 |
| | |
| **Total liabilities and equity** | **41,300** |

# Breaking Down the Balance Sheet of a Business ($ Millions)

Most liquid ↓ Least liquid

| Assets | | Liabilities and equity | |
|---|---|---|---|
| Current assets | | Current liabilities | |
| Cash and marketable sec. | 2,600 | Debt due for repayment | 200 |
| Accounts receivable | 500 | Accounts payable | 9,900 |
| Inventories | 9,000 | Total current liabilities | 10,100 |
| Total current assets | 12,100 | Long-term liabilities | 11,300 |
| Fixed assets | | Total liabilities | 21,400 |
| Tangible fixed assets | | | |
| Property, plant, equipment | 48,200 | Shareholders' equity | |
| Less accum. depreciation | 19,700 | Paid-in capital | 6,300 |
| Net tangible fixed assets | 28,500 | Retained earnings | 13,600 |
| Intangible asset | 700 | Total shareholders' equity | 19,900 |
| Total fixed assets | 29,200 | | |
| **Total assets** | **41,300** | **Total liabilities and equity** | **41,300** |

Paid first ↓ Paid last

- **Liquidity**, the ease of converting an asset into cash

# Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
    - Values on balance sheets are book value

# Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
   - Values on balance sheets are book value
   - E.g., the $2 billion data center built last year $\rightarrow$ book value, forever

# Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
   - Values on balance sheets are book value
   - E.g., the $2 billion data center built last year $\rightarrow$ book value, forever
2. **Market values**, what things would sell for today
   - E.g., worth $4 billion today $\rightarrow$ market value

# Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
   - Values on balance sheets are book value
   - E.g., the $2 billion data center built last year $\rightarrow$ book value, forever
2. **Market values**, what things would sell for today
   - E.g., worth $4 billion today $\rightarrow$ market value

- In general, book $\neq$ market value

# Example, Issuing Stock to Build a Data Center ($ Billions)

| 1. Today | | | |
|---|---|---|---|
| Cash | 1 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 1 |
| **Total assets** | **1** | **Total liabilities and equity** | **1** |

| 2. Issue stock | | | |
|---|---|---|---|
| Cash | 2 | Liabilities | 0 |
| Fixed assets | 0 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

| 3. Build data center | | | |
|---|---|---|---|
| Cash | 0 | Liabilities | 0 |
| Fixed assets | 2 | Shareholders' equity | 2 |
| **Total assets** | **2** | **Total liabilities and equity** | **2** |

- *Should you revise fixed assets if the data center is now worth more than $2 billion?* **No**
- *Should you revise shareholders' equity if someone is willing to buy the business at $4 billion?* **No**

# Book Value vs. Market Value

1. Book value of equity = Book value of assets − Book value of liabilities
   - E.g., Apple at the end of fiscal year 2025,
     - Book value of assets, $359,241 Mn
     - Book value of liabilities, $285,508 Mn
     - What is the book value of equity?

# Book Value vs. Market Value

1. Book value of equity = Book value of assets − Book value of liabilities
   - E.g., Apple at the end of fiscal year 2025,
     - Book value of assets, $359,241 Mn
     - Book value of liabilities, $285,508 Mn
     - What is the book value of equity?
     - Answer, $359,241 Mn − $285,508 Mn = $73,733 Mn

# Book Value vs. Market Value

1. Book value of equity = Book value of assets − Book value of liabilities
   - E.g., Apple at the end of fiscal year 2025,
     - Book value of assets, $359,241 Mn
     - Book value of liabilities, $285,508 Mn
     - What is the book value of equity?
     - Answer, $359,241 Mn − $285,508 Mn = $73,733 Mn

2. Market value of equity = Market price of each share × Total shares outstanding
   - E.g., Apple at the end of 2025,
     - Share price, about $271
     - Shares outstanding, 14,800 Mn
     - What is the market value of equity?

# Book Value vs. Market Value

1. Book value of equity = Book value of assets − Book value of liabilities
   - E.g., Apple at the end of fiscal year 2025,
     - Book value of assets, \$359,241 Mn
     - Book value of liabilities, \$285,508 Mn
     - What is the book value of equity?
     - Answer, \$359,241 Mn − \$285,508 Mn = \$73,733 Mn

2. Market value of equity = Market price of each share × Total shares outstanding
   - E.g., Apple at the end of 2025,
     - Share price, about \$271
     - Shares outstanding, 14,800 Mn
     - What is the market value of equity?
     - Answer, \$271 × 14,800 Mn = \$4,010,800 Mn

# Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm



# Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm
- Market value of equity reflects the money investors expect the firm to make in the future



# Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm
- Market value of equity reflects the money investors expect the firm to make in the future
- Naturally, market value of equity $>$ book value of equity for a successful business



# Practice Problem

- A business's book value of total assets is $45,000. Its book value of current liabilities is $15,000, and its book value of shareholders' equity is $20,000. How much is its book value of long-term liabilities?

# Practice Problem

- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?
- Assets = Liabilities + Equity

# Practice Problem

- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?
- Assets = Liabilities + Equity
- \$45,000 = \$15,000 + _____ + \$20,000

# Practice Problem

- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?
- Assets = Liabilities + Equity
- \$45,000 = \$15,000 + _____ + \$20,000
- \$45,000 = \$15,000 + \$10,000 + \$20,000

# Practice Problem

- Which of the following are current assets? Select all that apply.

a. Cash.
b. A software license (3-year term).
c. Wages payable (due in 2 weeks).
d. A bank loan (due in 3 years).
e. Accounts receivable.
f. An office building.

# Practice Problem

- Which of the following are current assets? Select all that apply.

a. Cash.
b. A software license (3-year term).
c. Wages payable (due in 2 weeks).
d. A bank loan (due in 3 years).
e. Accounts receivable.
f. An office building.

# Recap of Part 1

- Accounting conveys information to investors and stakeholders
- The balance sheet, assets, liabilities, and equity at a point in time
- Assets $=$ Liabilities $+$ Equity
- In general, book $\neq$ market value

## Module 2, Part 2

# The Income Statement

# Income Statement

- **Income statement**, how much firms earned in the past year



INCOME STATEMENTS

# Income Statement

- **Revenues**, sales made in the period, whether or not cash was collected
  - E.g., a firm sells 100 products to customers, each at a price of \$100
  - Revenues $= 100 \times \$100 =$ \$10,000

| | \$ Million |
|---|---|
| Revenues | 80,000 |
| COGS | 56,000 |
| SG&A expenses | 16,000 |
| Depreciation | 3,000 |
| **EBIT** | **5,000** |
| Interest expense | 1,000 |
| **Taxable income** | **4,000** |
| Taxes | 840 |
| **Net income** | **3,160** |
| Allocation of net income | |
| Dividends | 1,000 |
| Addition to retained earnings | 2,160 |

# Income Statement

- **Operating expenses**, the costs of running the business

| | $ Million |
|---|---|
| Revenues | 80,000 |
| COGS | 56,000 |
| SG&A expenses | 16,000 |
| Depreciation | 3,000 |
| **EBIT** | **5,000** |
| Interest expense | 1,000 |
| **Taxable income** | **4,000** |
| Taxes | 840 |
| **Net income** | **3,160** |
| Allocation of net income | |
| Dividends | 1,000 |
| Addition to retained earnings | 2,160 |

# Income Statement

- **Operating expenses**, the costs of running the business
  - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
    - E.g., each of the 100 products sold costs the firm \$60 to make
    - COGS $= 100 \times \$60 =$ \$6,000

| | $ Million |
|---|---|
| Revenues | 80,000 |
| COGS | 56,000 |
| SG&A expenses | 16,000 |
| Depreciation | 3,000 |
| **EBIT** | **5,000** |
| Interest expense | 1,000 |
| **Taxable income** | **4,000** |
| Taxes | 840 |
| **Net income** | **3,160** |
| Allocation of net income | |
| Dividends | 1,000 |
| Addition to retained earnings | 2,160 |

# Income Statement

- **Operating expenses**, the costs of running the business
  - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
    - E.g., each of the 100 products sold costs the firm \$60 to make
    - COGS $= 100 \times \$60 =$ \$6,000
  - **Selling, general & administrative (SG&A)**, other operating costs, e.g., marketing, salaries, distribution

| | $ Million |
|---|---|
| Revenues | 80,000 |
| COGS | 56,000 |
| SG&A expenses | 16,000 |
| Depreciation | 3,000 |
| **EBIT** | **5,000** |
| Interest expense | 1,000 |
| **Taxable income** | **4,000** |
| Taxes | 840 |
| **Net income** | **3,160** |
| Allocation of net income | |
| Dividends | 1,000 |
| Addition to retained earnings | 2,160 |

# Income Statement

- **Operating expenses**, the costs of running the business
  - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
    - E.g., each of the 100 products sold costs the firm \$60 to make
    - COGS $= 100 \times \$60 =$ \$6,000
  - **Selling, general & administrative (SG&A)**, other operating costs, e.g., marketing, salaries, distribution
  - **Depreciation**, this period's share of a fixed asset's cost

| | $ Million |
|---|---|
| Revenues | 80,000 |
| COGS | 56,000 |
| SG&A expenses | 16,000 |
| Depreciation | 3,000 |
| **EBIT** | **5,000** |
| Interest expense | 1,000 |
| **Taxable income** | **4,000** |
| Taxes | 840 |
| **Net income** | **3,160** |
| Allocation of net income | |
| Dividends | 1,000 |
| Addition to retained earnings | 2,160 |

# Fixed Assets Spread over Time as Depreciation

- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once



# Fixed Assets Spread over Time as Depreciation

- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once
- This spreading is **depreciation**



# Fixed Assets Spread over Time as Depreciation

- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once
- This spreading is **depreciation**
- Two things happen with depreciation,
  - In the income statement, subtract depreciation from revenues as a non-cash expense



# Fixed Assets Spread over Time as Depreciation

- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once
- This spreading is **depreciation**
- Two things happen with depreciation,
  - In the income statement, subtract depreciation from revenues as a non-cash expense
  - In the **balance sheet**, add to accumulated depreciation



# Fixed Assets Spread over Time as Depreciation

- The firm pays $2 billion to build a data center that lasts 5 years

# Fixed Assets Spread over Time as Depreciation

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years

# Fixed Assets Spread over Time as Depreciation

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is $0.4 billion

# Fixed Assets Spread over Time as Depreciation

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is $0.4 billion

| Year | 0 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|
| **Cash flow ($ billions)** | **−2.0** | 0 | 0 | 0 | 0 | 0 | |
| **Effect on net income** | 0 | **−0.4** | **−0.4** | **−0.4** | **−0.4** | **−0.4** | ← Depreciation |

# Income Statement

- **EBIT**, earnings before interest and taxes
  - EBIT = Revenues − COGS − SG&A expenses − Depreciation

| | $ Million |
|---|---|
| Revenues | 80,000 |
| COGS | 56,000 |
| SG&A expenses | 16,000 |
| Depreciation | 3,000 |
| **EBIT** | **5,000** |
| Interest expense | 1,000 |
| **Taxable income** | **4,000** |
| Taxes | 840 |
| **Net income** | **3,160** |
| Allocation of net income | |
| Dividends | 1,000 |
| Addition to retained earnings | 2,160 |

# Income Statement

- **Interest expense**, the interest paid on the firm's debt for the period

| | $ Million |
|---|---|
| Revenues | 80,000 |
| COGS | 56,000 |
| SG&A expenses | 16,000 |
| Depreciation | 3,000 |
| **EBIT** | **5,000** |
| Interest expense | 1,000 |
| **Taxable income** | **4,000** |
| Taxes | 840 |
| **Net income** | **3,160** |
| Allocation of net income | |
| Dividends | 1,000 |
| Addition to retained earnings | 2,160 |

# Income Statement

- **Taxes**
  - Taxable income = EBIT − Interest expense
  - Taxes = Taxable income × Tax rate

| | $ Million |
|---|---|
| Revenues | 80,000 |
| COGS | 56,000 |
| SG&A expenses | 16,000 |
| Depreciation | 3,000 |
| **EBIT** | **5,000** |
| Interest expense | 1,000 |
| **Taxable income** | **4,000** |
| Taxes | 840 |
| **Net income** | **3,160** |
| Allocation of net income | |
| Dividends | 1,000 |
| Addition to retained earnings | 2,160 |

# Income Statement

- **Net income**, what the business earned in the period after every expense

| | $ Million |
|---|---|
| Revenues | 80,000 |
| COGS | 56,000 |
| SG&A expenses | 16,000 |
| Depreciation | 3,000 |
| **EBIT** | **5,000** |
| Interest expense | 1,000 |
| **Taxable income** | **4,000** |
| Taxes | 840 |
| **Net income** | **3,160** |
| Allocation of net income | |
| Dividends | 1,000 |
| Addition to retained earnings | 2,160 |

# Where Net Income Goes

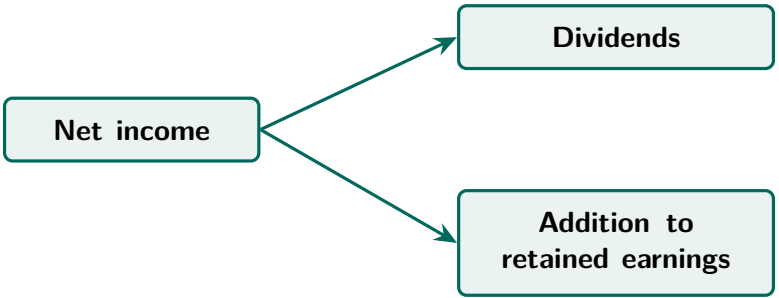
- Shareholders have claims to net income

# Where Net Income Goes

- Shareholders have claims to net income
- Net income can be kept in the business, or paid out to shareholders

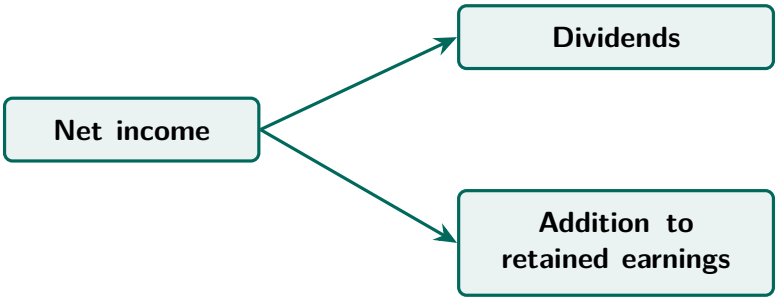
# Where Net Income Goes

- Shareholders have claims to net income
- Net income can be kept in the business, or paid out to shareholders



# Where Net Income Goes

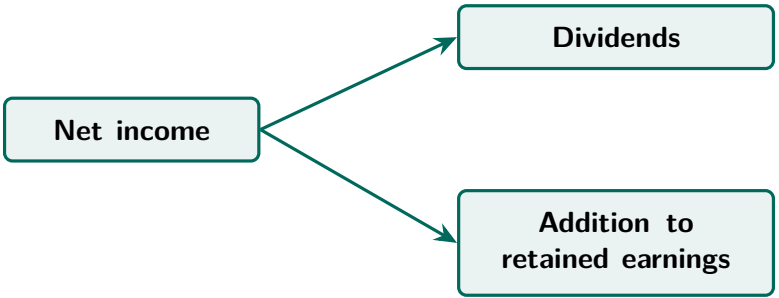
- Shareholders have claims to net income
- Net income can be kept in the business, or paid out to shareholders




- **Dividends**, cash paid out to shareholders now

# Where Net Income Goes

- Shareholders have claims to net income
- Net income can be kept in the business, or paid out to shareholders




- **Dividends**, cash paid out to shareholders now
- **Retained earnings**, accumulated profits kept in the business and reinvested on the shareholders' behalf

# Income Statement, A Memory Aid

| Line item | Letters | Remember it as |
|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
| SG&A expenses | S | **S**ells |
| Depreciation | D | **D**ependably |
| EBIT | E | **E**specially |
| Interest expense | I | **I**n |
| Taxable income | T.I. | **T**exas **I**ncorporated |
| Taxes | T | **T**exas |
| Net income | N.I. | **N**ever **I**ncorporates |
| Dividends | D | **D**uring |
| Addition to retained earnings | R.E. | **R**e-**E**lections |

# Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?




CNBC, August 18, 2026 (3:50), "Anthropic tells investors annualized revenue run rate climbed to $65 billion in July"

# Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- Profitable on an **EBITDA** basis
  - EBITDA, earnings before interest, taxes, depreciation, and amortization
  - Amortization, the same cost-spreading as depreciation, applied to intangible assets




CNBC, August 18, 2026 (3:50), "Anthropic tells investors annualized revenue run rate climbed to $65 billion in July"

# Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- Profitable on an **EBITDA** basis
  - EBITDA, earnings before interest, taxes, depreciation, and amortization
  - Amortization, the same cost-spreading as depreciation, applied to intangible assets
- Not profitable on a **net income** basis




CNBC, August 18, 2026 (3:50), "Anthropic tells investors annualized revenue run rate climbed to $65 billion in July"

# Practice Problem

| | $ |
|---|---|
| Revenues | ? |
| COGS | 20,000 |
| SG&A expenses | 10,000 |
| Depreciation | 8,000 |
| **EBIT** | **12,000** |
| Interest expense | ? |
| **Taxable income** | **?** |
| Taxes (tax rate = 21%) | 2,100 |
| **Net income** | **7,900** |
| Dividends | 2,500 |
| Addition to retained earnings | ? |

# Practice Problem

| | $ |
|---|---|
| Revenues | 50,000 |
| COGS | 20,000 |
| SG&A expenses | 10,000 |
| Depreciation | 8,000 |
| **EBIT** | **12,000** |
| Interest expense | 2,000 |
| **Taxable income** | **10,000** |
| Taxes (tax rate = 21%) | 2,100 |
| **Net income** | **7,900** |
| Dividends | 2,500 |
| Addition to retained earnings | 5,400 |

# Recap of Part 2

- The income statement, how much firms earned in the past year
- **Depreciation**, this period's share of a fixed asset's cost
- Net income can be kept in the business, or paid out to shareholders
- Taxes are computed **after** depreciation and interest, never before

## Module 2, Part 3

# The Statement of Cash Flows

# Statement of Cash Flows

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made

# Statement of Cash Flows

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period

# Statement of Cash Flows

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period
- The statement has three parts
  - **Operating activities**, cash from running the business day to day

# Statement of Cash Flows

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period
- The statement has three parts
  - **Operating activities**, cash from running the business day to day
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them

# Statement of Cash Flows

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period
- The statement has three parts
  - **Operating activities**, cash from running the business day to day
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

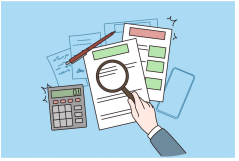
# Statement of Cash Flows

| Direct method | Indirect method |
|---|---|
| Lists every cash receipt and payment | Starts from net income and adjusts it |
| More time-consuming to prepare | Faster to prepare |
| Rarely used in practice | **The common choice** |

**Direct** →



**Indirect** →



# Statement of Cash Flows

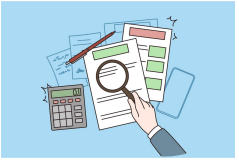
| Direct method | Indirect method |
|---|---|
| Lists every cash receipt and payment | Starts from net income and adjusts it |
| More time-consuming to prepare | Faster to prepare |
| Rarely used in practice | **The common choice** |

- Both methods give the same totals

**Direct** →



**Indirect** →



# Statement of Cash Flows

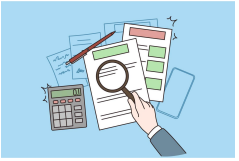
| Direct method | Indirect method |
|---|---|
| Lists every cash receipt and payment | Starts from net income and adjusts it |
| More time-consuming to prepare | Faster to prepare |
| Rarely used in practice | **The common choice** |

- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

**Direct** →



**Indirect** →



# How Do Net Income and Cash Flow Differ?

- The firm pays $2 billion to build a data center that lasts 5 years

# How Do Net Income and Cash Flow Differ?

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years

# How Do Net Income and Cash Flow Differ?

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is $0.4 billion

# How Do Net Income and Cash Flow Differ?

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is $0.4 billion

| Year | 0 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|
| **Cash flow ($ billions)** | **−2.0** | 0 | 0 | 0 | 0 | 0 | |
| **Effect on net income** | 0 | **−0.4** | **−0.4** | **−0.4** | **−0.4** | **−0.4** | ← Depreciation |

# How Do Net Income and Cash Flow Differ?

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is $0.4 billion

| Year | 0 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|
| **Cash flow ($ billions)** | **−2.0** | 0 | 0 | 0 | 0 | 0 | |
| **Effect on net income** | 0 | **−0.4** | **−0.4** | **−0.4** | **−0.4** | **−0.4** | ← Depreciation |

- Cash leaves all at once, net income feels it slowly

# How Do Net Income and Cash Flow Differ?

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is $0.4 billion

| Year | 0 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|
| **Cash flow ($ billions)** | **−2.0** | 0 | 0 | 0 | 0 | 0 | |
| **Effect on net income** | 0 | **−0.4** | **−0.4** | **−0.4** | **−0.4** | **−0.4** | ← Depreciation |

- Cash leaves all at once, net income feels it slowly

***Relationship of cash flow (CF) and net income (NI)?***

# How Do Net Income and Cash Flow Differ?

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is $0.4 billion

| Year | 0 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|
| **Cash flow ($ billions)** | **−2.0** | 0 | 0 | 0 | 0 | 0 | |
| **Effect on net income** | 0 | **−0.4** | **−0.4** | **−0.4** | **−0.4** | **−0.4** | ← Depreciation |

- Cash leaves all at once, net income feels it slowly

***Relationship of cash flow (CF) and net income (NI)?***

***When spending on fixed assets, CF = NI − spending on fixed assets***

# How Do Net Income and Cash Flow Differ?

- The firm pays $2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is $0.4 billion

| Year | 0 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|
| **Cash flow ($ billions)** | **−2.0** | 0 | 0 | 0 | 0 | 0 | |
| **Effect on net income** | 0 | **−0.4** | **−0.4** | **−0.4** | **−0.4** | **−0.4** | ← Depreciation |

- Cash leaves all at once, net income feels it slowly

***Relationship of cash flow (CF) and net income (NI)?***

***When spending on fixed assets, CF = NI − spending on fixed assets***

***When recording depreciation, CF = NI + depreciation***

# How Do Net Income and Cash Flow Differ?

- The firm issues (sells) $1 billion of new stock to investors

# How Do Net Income and Cash Flow Differ?

- The firm issues (sells) $1 billion of new stock to investors

| | Effect |
|---|---|
| **Cash flow ($ billions)** | **+1.0** |
| **Effect on net income** | **0** |

# How Do Net Income and Cash Flow Differ?

- The firm issues (sells) $1 billion of new stock to investors

| | Effect |
|---|---|
| **Cash flow ($ billions)** | **+1.0** |
| **Effect on net income** | **0** |

- Selling stock is not generating income

# How Do Net Income and Cash Flow Differ?

- The firm issues (sells) $1 billion of new stock to investors

| | Effect |
|---|---|
| **Cash flow ($ billions)** | **+1.0** |
| **Effect on net income** | **0** |

- Selling stock is not generating income
- Net income never records money raised from issuing (selling) stocks or bonds or borrowing from banks

# How Do Net Income and Cash Flow Differ?

- The firm issues (sells) $1 billion of new stock to investors

| | Effect |
|---|---|
| **Cash flow ($ billions)** | **+1.0** |
| **Effect on net income** | **0** |

- Selling stock is not generating income
- Net income never records money raised from issuing (selling) stocks or bonds or borrowing from banks

***When raising money, CF = NI + amount raised***

# How Do Net Income and Cash Flow Differ?

- The firm spends $60 million producing AI control panels
- None of them have been sold yet




OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm spends $60 million producing AI control panels
- None of them have been sold yet

| | Effect |
|---|---|
| **Cash flow ($ millions)** | **−60** |
| **Effect on net income** | **0** |
| **Change in inventories** | **+60** |




OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm spends $60 million producing AI control panels
- None of them have been sold yet

| | Effect |
|---|---|
| **Cash flow ($ millions)** | **−60** |
| **Effect on net income** | **0** |
| **Change in inventories** | **+60** |

- The income statement records revenues and cost of goods sold (COGS) at the time of sale




OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet

| | Effect |
|---|---|
| **Cash flow (\$ millions)** | **−60** |
| **Effect on net income** | **0** |
| **Change in inventories** | **+60** |

- The income statement records revenues and cost of goods sold (COGS) at the time of sale
- COGS is subtracted only when the panels are sold, not when they are produced




OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet

| | Effect |
|---|---|
| **Cash flow (\$ millions)** | **−60** |
| **Effect on net income** | **0** |
| **Change in inventories** | **+60** |

- The income statement records revenues and cost of goods sold (COGS) at the time of sale
- COGS is subtracted only when the panels are sold, not when they are produced




OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

***When adding to inventories, CF = NI − change in inventories***

# How Do Net Income and Cash Flow Differ?

- The firm sells AI control panels for $100 million
- The customer has not paid yet




OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm sells AI control panels for $100 million
- The customer has not paid yet

| | Effect |
|---|---|
| **Cash flow ($ millions)** | **0** |
| **Effect on net income** | **+100** |
| **Change in accounts receivable** | **+100** |




OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm sells AI control panels for $100 million
- The customer has not paid yet

| | Effect |
|---|---|
| **Cash flow ($ millions)** | **0** |
| **Effect on net income** | **+100** |
| **Change in accounts receivable** | **+100** |

- Net income rises now, cash flow comes later when customers actually pay




OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm sells AI control panels for $100 million
- The customer has not paid yet

| | Effect |
|---|---|
| **Cash flow ($ millions)** | **0** |
| **Effect on net income** | **+100** |
| **Change in accounts receivable** | **+100** |

- Net income rises now, cash flow comes later when customers actually pay




OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

***When firm not yet paid by customers, CF = NI − change in accounts receivable***

# How Do Net Income and Cash Flow Differ?

- The firm owed suppliers $30 million for electronic parts
- The firm pays down the $30 million




OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm owed suppliers \$30 million for electronic parts
- The firm pays down the \$30 million

| | Effect |
|---|---|
| **Cash flow (\$ millions)** | **−30** |
| **Effect on net income** | **0** |
| **Change in accounts payable** | **−30** |




OpenAI, July 15, 2026 (2:15),
"Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm owed suppliers $30 million for electronic parts
- The firm pays down the $30 million

| | Effect |
|---|---|
| **Cash flow ($ millions)** | **−30** |
| **Effect on net income** | **0** |
| **Change in accounts payable** | **−30** |

- No sale of panels happened. No revenues or COGS should be recorded. Net income is 0




OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm owed suppliers $30 million for electronic parts
- The firm pays down the $30 million

| | Effect |
|---|---|
| **Cash flow ($ millions)** | **−30** |
| **Effect on net income** | **0** |
| **Change in accounts payable** | **−30** |

- No sale of panels happened. No revenues or COGS should be recorded. Net income is 0




OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

# How Do Net Income and Cash Flow Differ?

- The firm owed suppliers $30 million for electronic parts
- The firm pays down the $30 million

| | Effect |
|---|---|
| **Cash flow ($ millions)** | **−30** |
| **Effect on net income** | **0** |
| **Change in accounts payable** | **−30** |

- No sale of panels happened. No revenues or COGS should be recorded. Net income is 0




OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

***When firm pays off suppliers, CF = NI + change in accounts payable***

# Operating Activities

**Operating activities**, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
   - NWC = inventories + accounts receivable − accounts payable
   - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

# Why Subtract Change in Net Working Capital (NWC)?

*When adding to inventories, CF = NI − change in inventories*

*When firm not yet paid by customers, CF = NI − change in accounts receivable*

*When firm pays off suppliers, CF = NI + change in accounts payable*

# Why Subtract Change in Net Working Capital (NWC)?

*When adding to inventories, CF = NI − change in inventories*

*When firm not yet paid by customers, CF = NI − change in accounts receivable*

*When firm pays off suppliers, CF = NI + change in accounts payable*

*Ignoring other adjustments,*

# Why Subtract Change in Net Working Capital (NWC)?

*When adding to inventories, CF = NI − change in inventories*

*When firm not yet paid by customers, CF = NI − change in accounts receivable*

*When firm pays off suppliers, CF = NI + change in accounts payable*

*Ignoring other adjustments,*

$$
\begin{aligned}
CF = NI \\
&- \text{change in inventories} \\
&- \text{change in accounts receivable} \\
&+ \text{change in accounts payable}
\end{aligned}
$$

# Why Subtract Change in Net Working Capital (NWC)?

*When adding to inventories, CF = NI − change in inventories*

*When firm not yet paid by customers, CF = NI − change in accounts receivable*

*When firm pays off suppliers, CF = NI + change in accounts payable*

*Ignoring other adjustments,*

$$\begin{aligned} CF = NI \\ &- \text{change in inventories} \\ &- \text{change in accounts receivable} \\ &+ \text{change in accounts payable} \end{aligned}$$

$$CF = NI - \text{change in (inventories} + \text{accounts receivable} - \text{accounts payable)}$$

# Why Subtract Change in Net Working Capital (NWC)?

*When adding to inventories, CF = NI − change in inventories*

*When firm not yet paid by customers, CF = NI − change in accounts receivable*

*When firm pays off suppliers, CF = NI + change in accounts payable*

*Ignoring other adjustments,*

$$\begin{aligned} CF = NI \\ &- \text{change in inventories} \\ &- \text{change in accounts receivable} \\ &+ \text{change in accounts payable} \end{aligned}$$

$$CF = NI - \text{change in (inventories} + \text{accounts receivable} - \text{accounts payable)}$$

$$CF = NI - \text{change in net working capital (NWC)}$$

# Operating Activities

| Cash flow from operating activities | |
|---|---|
| Net income | 3,160 |
| Depreciation | 3,000 |
| Change in net working capital | −1,000 |
| Cash flow from operating activities | 7,160 |

$$3{,}160 + 3{,}000 - (-1{,}000) = 7{,}160$$

# Investing Activities

**Investing activities**, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold
   - **Capital**, a firm's fixed assets
   - **Capital expenditure**, the amount spent on buying capital during the period
   - Cash flow is *negative* if firms buy capital

# Investing Activities

| Cash flow from investing activities | |
|---|---|
| Cash flow from capital expenditure | −3,000 |
| Cash flow from sales of fixed assets | 200 |
| Cash flow from investing activities | −2,800 |

$$-3{,}000 + 200 = -2{,}800$$

# Financing Activities

**Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Financing activities show cash flows from,

1. Selling new shares $\rightarrow$ Cash $\uparrow$
2. Paying dividends $\rightarrow$ Cash $\downarrow$
3. Taking new debt, either a loan or a bond $\rightarrow$ Cash $\uparrow$
4. Repaying old debt $\rightarrow$ Cash $\downarrow$

# Practice Problem

- During the year, a business sells new shares for $120,000 and pays dividends of $30,000. It takes a new loan of $80,000 and repays $50,000 of old debt. How much is its cash flow from financing activities?

# Practice Problem

- During the year, a business sells new shares for \$120,000 and pays dividends of \$30,000. It takes a new loan of \$80,000 and repays \$50,000 of old debt. How much is its cash flow from financing activities?
- Cash flow from financing activities $= \$120{,}000 - \$30{,}000 + \$80{,}000 - \$50{,}000 = \$120{,}000$

# Recap of Part 3

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
  - **Operating activities**, cash flow = Net income + Depreciation − Change in NWC
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

# Practice Problem

- Which of the following will make cash flow higher than net income for a firm?

a. Increase in inventories.
b. Decrease in accounts payable.
c. Decrease in accounts receivable.
d. Expenditures on fixed assets.

# Practice Problem

- Which of the following will make cash flow higher than net income for a firm?

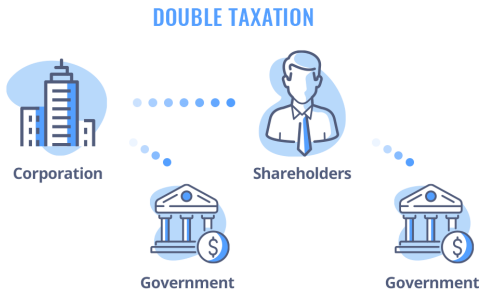
a. Increase in inventories.
b. Decrease in accounts payable.
c. Decrease in accounts receivable.
d. Expenditures on fixed assets.

## Module 2, Part 4

# Taxation

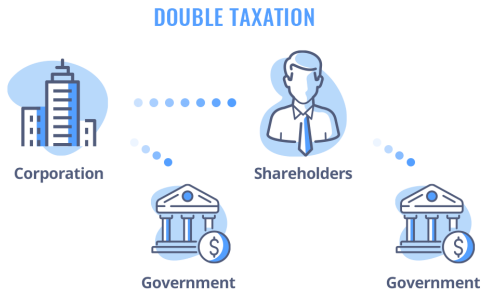
# Corporate Taxation, Double Taxation

- Recall from **Module 1**,
  - The corporation pays tax on its income



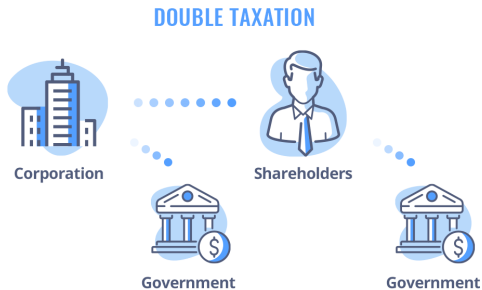
# Corporate Taxation, Double Taxation

- Recall from **Module 1**,
  - The corporation pays tax on its income
  - Shareholders pay tax again on what they receive



# Corporate Taxation, Double Taxation

- Recall from **Module 1**,
  - The corporation pays tax on its income
  - Shareholders pay tax again on what they receive
- This is **double taxation**



# Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**




WSJ, February 28, 2025 (6:57),
"Economists on How Trump's 2017 Tax Cuts Actually Played Out"

# Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**
- The rate is a major political topic




WSJ, February 28, 2025 (6:57),
"Economists on How Trump's 2017 Tax Cuts Actually Played Out"

# Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**
- The rate is a major political topic
- **Tax Cuts and Jobs Act (TCJA, 2017)**, cut the rate from 35% to 21%




WSJ, February 28, 2025 (6:57), "Economists on How Trump's 2017 Tax Cuts Actually Played Out"

# Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**
- The rate is a major political topic
- **Tax Cuts and Jobs Act (TCJA, 2017)**, cut the rate from 35% to 21%
- The tax cut was extended and largely made permanent in 2025




WSJ, February 28, 2025 (6:57), "Economists on How Trump's 2017 Tax Cuts Actually Played Out"

# Tax Shield

- **Tax shield**, the reduction in the taxes a corporation needs to pay that results from a tax-deductible expense



# Tax Shield

- **Tax shield**, the reduction in the taxes a corporation needs to pay that results from a tax-deductible expense
- **Tax-deductible expense**, an expense on the income statement that comes before taxable income



# Interest Tax Shield

- Interest comes first

| Line item | Letters | Remember it as |
|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
| SG&A expenses | S | **S**ells |
| Depreciation | D | **D**ependably |
| EBIT | E | **E**specially |
| Interest expense | I | **I**n |
| Taxable income | T.I. | **T**exas **I**ncorporated |
| Taxes | T | **T**exas |
| Net income | N.I. | **N**ever **I**ncorporates |
| Dividends | D | **D**uring |
| Addition to retained earnings | R.E. | **R**e-**E**lections |

# Interest Tax Shield

- Interest comes first
- Taxes come last

| Line item | Letters | Remember it as |
|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
| SG&A expenses | S | **S**ells |
| Depreciation | D | **D**ependably |
| EBIT | E | **E**specially |
| Interest expense | I | **I**n |
| Taxable income | T.I. | **T**exas **I**ncorporated |
| Taxes | T | **T**exas |
| Net income | N.I. | **N**ever **I**ncorporates |
| Dividends | D | **D**uring |
| Addition to retained earnings | R.E. | **R**e-**E**lections |

# Interest Tax Shield

- Interest comes first
- Taxes come last
- Interest reduces corporate taxes

| Line item | Letters | Remember it as |
|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
| SG&A expenses | S | **S**ells |
| Depreciation | D | **D**ependably |
| EBIT | E | **E**specially |
| Interest expense | I | **I**n |
| Taxable income | T.I. | **T**exas **I**ncorporated |
| Taxes | T | **T**exas |
| Net income | N.I. | **N**ever **I**ncorporates |
| Dividends | D | **D**uring |
| Addition to retained earnings | R.E. | **R**e-**E**lections |

# Interest Tax Shield

- Interest expense can
  - act as a “shield” against and reduce corporate taxation

# Interest Tax Shield

- Interest expense can
  - act as a "shield" against and reduce corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)

# Interest Tax Shield, Example

| Corporation A | |
|---|---|
| **EBIT** | **$5,000** |
| Interest expense | $1,000 |
| **Taxable income** | **?** |
| Taxes (tax rate = 21%) | ? |
| **Net income** | **?** |

| Corporation B | |
|---|---|
| **EBIT** | **$5,000** |
| Interest expense | $0 |
| **Taxable income** | **?** |
| Taxes (tax rate = 21%) | ? |
| **Net income** | **?** |

# Interest Tax Shield, Example

| Corporation A | |
|---|---|
| **EBIT** | **$5,000** |
| Interest expense | $1,000 |
| **Taxable income** | **$4,000** |
| Taxes (tax rate = 21%) | $840 |
| **Net income** | **$3,160** |

| Corporation B | |
|---|---|
| **EBIT** | **$5,000** |
| Interest expense | $0 |
| **Taxable income** | **?** |
| Taxes (tax rate = 21%) | ? |
| **Net income** | **?** |

# Interest Tax Shield, Example

| Corporation A | |
|---|---|
| **EBIT** | **$5,000** |
| Interest expense | $1,000 |
| **Taxable income** | **$4,000** |
| Taxes (tax rate = 21%) | $840 |
| **Net income** | **$3,160** |

| Corporation B | |
|---|---|
| **EBIT** | **$5,000** |
| Interest expense | $0 |
| **Taxable income** | **$5,000** |
| Taxes (tax rate = 21%) | $1,050 |
| **Net income** | **$3,950** |

# Interest Tax Shield, Example

| Corporation A | | |
|---|---|---|
| **EBIT** | **$5,000** | |
| Interest expense | $1,000 | → debt investors |
| **Taxable income** | **$4,000** | |
| Taxes (tax rate = 21%) | $840 | |
| **Net income** | **$3,160** | |

| Corporation B | | |
|---|---|---|
| **EBIT** | **$5,000** | |
| Interest expense | $0 | → debt investors |
| **Taxable income** | **$5,000** | |
| Taxes (tax rate = 21%) | $1,050 | |
| **Net income** | **$3,950** | |

# Interest Tax Shield, Example

| Corporation A | | |
|---|---|---|
| **EBIT** | **$5,000** | |
| Interest expense | $1,000 | → debt investors |
| **Taxable income** | **$4,000** | |
| Taxes (tax rate = 21%) | $840 | |
| **Net income** | **$3,160** | → equity investors |

| Corporation B | | |
|---|---|---|
| **EBIT** | **$5,000** | |
| Interest expense | $0 | → debt investors |
| **Taxable income** | **$5,000** | |
| Taxes (tax rate = 21%) | $1,050 | |
| **Net income** | **$3,950** | → equity investors |

# Interest Tax Shield, Example

| Corporation A | | | |
|---|---|---|---|
| **EBIT** | **$5,000** | | |
| Interest expense | $1,000 | → debt investors | |
| **Taxable income** | **$4,000** | | |
| Taxes (tax rate = 21%) | $840 | | **$4,160** |
| **Net income** | **$3,160** | → equity investors | |

| Corporation B | | | |
|---|---|---|---|
| **EBIT** | **$5,000** | | |
| Interest expense | $0 | → debt investors | |
| **Taxable income** | **$5,000** | | |
| Taxes (tax rate = 21%) | $1,050 | | **$3,950** |
| **Net income** | **$3,950** | → equity investors | |

# Interest Tax Shield, Example

| Corporation A | | | |
|---|---|---|---|
| **EBIT** | **$5,000** | | |
| Interest expense | $1,000 | → debt investors | |
| **Taxable income** | **$4,000** | | |
| Taxes (tax rate = 21%) | $840 | | **$4,160** |
| **Net income** | **$3,160** | → equity investors | |

| Corporation B | | | |
|---|---|---|---|
| **EBIT** | **$5,000** | | |
| Interest expense | $0 | → debt investors | |
| **Taxable income** | **$5,000** | | |
| Taxes (tax rate = 21%) | $1,050 | | **$3,950** |
| **Net income** | **$3,950** | → equity investors | |

***Investors get $4,160 in total with debt, $3,950 without***

# Interest Tax Shield, Example

| Corporation A | | | |
|---|---|---|---|
| **EBIT** | **$5,000** | | |
| Interest expense | $1,000 | → debt investors | |
| **Taxable income** | **$4,000** | | |
| Taxes (tax rate = 21%) | $840 | | **$4,160** |
| **Net income** | **$3,160** | → equity investors | |

| Corporation B | | | |
|---|---|---|---|
| **EBIT** | **$5,000** | | |
| Interest expense | $0 | → debt investors | |
| **Taxable income** | **$5,000** | | |
| Taxes (tax rate = 21%) | $1,050 | | **$3,950** |
| **Net income** | **$3,950** | → equity investors | |

***Investors get $4,160 in total with debt, $3,950 without***

***Investors get $210 more in total with debt***

# Interest Tax Shield

- Interest expense can
  - act as a "shield" against and reduce corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)

- **Interest tax shield**, the tax saving that interest expense creates, as interest is deducted (subtracted) before taxes

# Interest Tax Shield

- Interest expense can
  - act as a "shield" against and reduce corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)

- **Interest tax shield**, the tax saving that interest expense creates, as interest is deducted (subtracted) before taxes

$$\textbf{Interest tax shield} = \textbf{Tax rate} \times \textbf{Interest expense}$$

# Interest Tax Shield

- Interest expense can
  - act as a "shield" against and reduce corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)

- **Interest tax shield**, the tax saving that interest expense creates, as interest is deducted (subtracted) before taxes

$$\textbf{Interest tax shield} = \textbf{Tax rate} \times \textbf{Interest expense}$$

- In the example, $21\% \times \$1{,}000 = \$210$

# Interest Tax Shield

- Interest expense can
  - act as a "shield" against and reduce corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)

- **Interest tax shield**, the tax saving that interest expense creates, as interest is deducted (subtracted) before taxes

$$\textbf{Interest tax shield} = \textbf{Tax rate} \times \textbf{Interest expense}$$

- In the example, $21\% \times \$1{,}000 = \$210$

- Should a corporation take on as much debt as possible? No, bankruptcy costs.

# Math for Interest Tax Shield

- Each dollar of interest lowers taxable income by $1

# Math for Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate (21%) $\times$ \$1 = ¢21

# Math for Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate (21%) $\times$ \$1 = ¢21

$$\underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by \$1}} - \underbrace{\text{Taxes}}_{\downarrow \text{ by ¢21}}$$

# Math for Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate (21%) $\times$ \$1 = ¢21

$$\underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by \$1}} - \underbrace{\text{Taxes}}_{\downarrow \text{ by ¢21}}$$

- Equity investors get ¢79 less

# Math for Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate (21%) $\times$ \$1 = ¢21

$$\underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by \$1}} - \underbrace{\text{Taxes}}_{\downarrow \text{ by ¢21}}$$

- Equity investors get ¢79 less
- Debt investors get \$1 more

# Math for Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate (21%) $\times$ \$1 = ¢21

$$\underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by \$1}} - \underbrace{\text{Taxes}}_{\downarrow \text{ by ¢21}}$$

- Equity investors get ¢79 less
- Debt investors get \$1 more

$$\underbrace{\text{Total CF to investors}}_{\uparrow \text{ by ¢21}} = \underbrace{\text{Debt investors}}_{\uparrow \text{ by \$1}} + \underbrace{\text{Equity investors}}_{\downarrow \text{ by ¢79}}$$

# Math for Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate (21%) $\times$ \$1 = ¢21

$$\underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by \$1}} - \underbrace{\text{Taxes}}_{\downarrow \text{ by ¢21}}$$

- Equity investors get ¢79 less
- Debt investors get \$1 more

$$\underbrace{\text{Total CF to investors}}_{\uparrow \text{ by ¢21}} = \underbrace{\text{Debt investors}}_{\uparrow \text{ by \$1}} + \underbrace{\text{Equity investors}}_{\downarrow \text{ by ¢79}}$$

- Each dollar of interest raises the *total* cash flow to investors by ¢21

# Practice Problem

- A corporation pays a 21% tax rate. It takes on debt and pays $1 more in interest each year. In that year, how does the cash flow to each investor group change?

a. Debt investors get $1 more, equity investors get $1 less, no change in total
b. Debt investors get $1 more, equity investors get $0.79 less, $0.21 more in total
c. Debt investors get $0.79 more, equity investors get $1 less, $0.21 less in total
d. Debt investors get $1 more, equity investors get $0.21 less, $0.79 more in total

# Practice Problem

- A corporation pays a 21% tax rate. It takes on debt and pays $1 more in interest each year. In that year, how does the cash flow to each investor group change?

a. Debt investors get $1 more, equity investors get $1 less, no change in total

b. Debt investors get $1 more, equity investors get $0.79 less, $0.21 more in total

c. Debt investors get $0.79 more, equity investors get $1 less, $0.21 less in total

d. Debt investors get $1 more, equity investors get $0.21 less, $0.79 more in total

- Net income falls by $(1 - 21\%) \times \$1 = \$0.79$, and the interest tax shield is $21\% \times \$1 = \$0.21$

# Depreciation Tax Shield

- Depreciation comes first

| Line item | Letters | Remember it as |
|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
| SG&A expenses | S | **S**ells |
| Depreciation | D | **D**ependably |
| EBIT | E | **E**specially |
| Interest expense | I | **I**n |
| Taxable income | T.I. | **T**exas **I**ncorporated |
| Taxes | T | **T**exas |
| Net income | N.I. | **N**ever **I**ncorporates |
| Dividends | D | **D**uring |
| Addition to retained earnings | R.E. | **R**e-**E**lections |

# Depreciation Tax Shield

- Depreciation comes first
- Taxes come last

| Line item | Letters | Remember it as |
|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
| SG&A expenses | S | **S**ells |
| Depreciation | D | **D**ependably |
| EBIT | E | **E**specially |
| Interest expense | I | **I**n |
| Taxable income | T.I. | **T**exas **I**ncorporated |
| Taxes | T | **T**exas |
| Net income | N.I. | **N**ever **I**ncorporates |
| Dividends | D | **D**uring |
| Addition to retained earnings | R.E. | **R**e-**E**lections |

# Depreciation Tax Shield

- Depreciation comes first
- Taxes come last
- Depreciation reduces corporate taxes

| Line item | Letters | Remember it as |
|---|---|---|
| Revenues | R | **R**ainbow |
| COGS | C | **C**andy |
| SG&A expenses | S | **S**ells |
| Depreciation | D | **D**ependably |
| EBIT | E | **E**specially |
| Interest expense | I | **I**n |
| Taxable income | T.I. | **T**exas **I**ncorporated |
| Taxes | T | **T**exas |
| Net income | N.I. | **N**ever **I**ncorporates |
| Dividends | D | **D**uring |
| Addition to retained earnings | R.E. | **R**e-**E**lections |

# Depreciation Tax Shield

- Depreciation can also
  - act as a “shield” against and reduce corporate taxation

# Depreciation Tax Shield

- Depreciation can also
  - act as a "shield" against and reduce corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)

# Depreciation Tax Shield, Example

- Example, Suppose a corporation has an *additional* $1,000 million in depreciation, and its tax rate is 21%

Income Statement ($ Million)

| | Baseline | |
|---|---|---|
| Revenues | 80,000 | |
| COGS | 56,000 | |
| SG&A expenses | 16,000 | |
| Depreciation | 3,000 | |
| **EBIT** | **5,000** | |
| Interest expense | 1,000 | |
| **Taxable income** | **4,000** | |
| Taxes | 840 | |
| **Net income** | **3,160** | |

# Depreciation Tax Shield, Example

- Example, Suppose a corporation has an *additional* $1,000 million in depreciation, and its tax rate is 21%

Income Statement ($ Million)

| | Baseline | More depreciation |
|---|---|---|
| Revenues | 80,000 | 80,000 |
| COGS | 56,000 | 56,000 |
| SG&A expenses | 16,000 | 16,000 |
| Depreciation | 3,000 | |
| **EBIT** | **5,000** | |
| Interest expense | 1,000 | |
| **Taxable income** | **4,000** | |
| Taxes | 840 | |
| **Net income** | **3,160** | |

# Depreciation Tax Shield, Example

- Example, Suppose a corporation has an *additional* $1,000 million in depreciation, and its tax rate is 21%

Income Statement ($ Million)

| | Baseline | More depreciation |
|---|---|---|
| Revenues | 80,000 | 80,000 |
| COGS | 56,000 | 56,000 |
| SG&A expenses | 16,000 | 16,000 |
| Depreciation | 3,000 | 4,000 |
| **EBIT** | **5,000** | |
| Interest expense | 1,000 | |
| **Taxable income** | **4,000** | |
| Taxes | 840 | |
| **Net income** | **3,160** | |

# Depreciation Tax Shield, Example

- Example, Suppose a corporation has an *additional* $1,000 million in depreciation, and its tax rate is 21%

Income Statement ($ Million)

| | Baseline | More depreciation |
|---|---|---|
| Revenues | 80,000 | 80,000 |
| COGS | 56,000 | 56,000 |
| SG&A expenses | 16,000 | 16,000 |
| Depreciation | 3,000 | 4,000 |
| **EBIT** | **5,000** | **4,000** |
| Interest expense | 1,000 | |
| **Taxable income** | **4,000** | |
| Taxes | 840 | |
| **Net income** | **3,160** | |

# Depreciation Tax Shield, Example

- Example, Suppose a corporation has an *additional* $1,000 million in depreciation, and its tax rate is 21%

Income Statement ($ Million)

| | Baseline | More depreciation |
|---|---|---|
| Revenues | 80,000 | 80,000 |
| COGS | 56,000 | 56,000 |
| SG&A expenses | 16,000 | 16,000 |
| Depreciation | 3,000 | 4,000 |
| **EBIT** | **5,000** | **4,000** |
| Interest expense | 1,000 | 1,000 |
| **Taxable income** | **4,000** | |
| Taxes | 840 | |
| **Net income** | **3,160** | |

# Depreciation Tax Shield, Example

- Example, Suppose a corporation has an *additional* $1,000 million in depreciation, and its tax rate is 21%

Income Statement ($ Million)

| | Baseline | More depreciation |
|---|---|---|
| Revenues | 80,000 | 80,000 |
| COGS | 56,000 | 56,000 |
| SG&A expenses | 16,000 | 16,000 |
| Depreciation | 3,000 | 4,000 |
| **EBIT** | **5,000** | **4,000** |
| Interest expense | 1,000 | 1,000 |
| **Taxable income** | **4,000** | **3,000** |
| Taxes | 840 | |
| **Net income** | **3,160** | |

# Depreciation Tax Shield, Example

- Example, Suppose a corporation has an *additional* $1,000 million in depreciation, and its tax rate is 21%

Income Statement ($ Million)

| | Baseline | More depreciation |
|---|---|---|
| Revenues | 80,000 | 80,000 |
| COGS | 56,000 | 56,000 |
| SG&A expenses | 16,000 | 16,000 |
| Depreciation | 3,000 | 4,000 |
| **EBIT** | **5,000** | **4,000** |
| Interest expense | 1,000 | 1,000 |
| **Taxable income** | **4,000** | **3,000** |
| Taxes | 840 | 630 |
| **Net income** | **3,160** | |

# Depreciation Tax Shield, Example

- Example, Suppose a corporation has an *additional* $1,000 million in depreciation, and its tax rate is 21%

Income Statement ($ Million)

| | Baseline | More depreciation |
|---|---|---|
| Revenues | 80,000 | 80,000 |
| COGS | 56,000 | 56,000 |
| SG&A expenses | 16,000 | 16,000 |
| Depreciation | 3,000 | 4,000 |
| **EBIT** | **5,000** | **4,000** |
| Interest expense | 1,000 | 1,000 |
| **Taxable income** | **4,000** | **3,000** |
| Taxes | 840 | 630 |
| **Net income** | **3,160** | **2,370** |

# Depreciation Tax Shield, Example

- Net income *falls* by \$790 million

| \$ Million | Baseline | More depreciation | Change |
|---|---|---|---|
| Net income | 3,160 | 2,370 | −790 |
| Add back depreciation | | | |
| **Cash flow** | | | |

# Depreciation Tax Shield, Example

- Net income *falls* by \$790 million
- Recall Part 3,
  CF from operating activities = NI + Depreciation − Change in NWC

| \$ Million | Baseline | More depreciation | Change |
|---|---|---|---|
| Net income | 3,160 | 2,370 | −790 |
| Add back depreciation | 3,000 | 4,000 | +1,000 |
| **Cash flow** | | | |

# Depreciation Tax Shield, Example

- Net income *falls* by $790 million
- Recall Part 3,
  CF from operating activities = NI + Depreciation − Change in NWC
- Equity investors have claims to cash flow from operating activities

| $ Million | Baseline | More depreciation | Change |
|---|---|---|---|
| Net income | 3,160 | 2,370 | −790 |
| Add back depreciation | 3,000 | 4,000 | +1,000 |
| **Cash flow** | | | |

# Depreciation Tax Shield, Example

- Net income *falls* by \$790 million
- Recall Part 3,
  CF from operating activities = NI + Depreciation − Change in NWC
- Equity investors have claims to cash flow from operating activities
- Cash flow *rises* by \$210 million

| \$ Million | Baseline | More depreciation | Change |
|---|---|---|---|
| Net income | 3,160 | 2,370 | −790 |
| Add back depreciation | 3,000 | 4,000 | +1,000 |
| **Cash flow** | **6,160** | **6,370** | **+210** |

# Depreciation Tax Shield

- Depreciation can also
  - act as a "shield" against and reduce corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)

- **Depreciation tax shield**, the tax saving that depreciation creates, as depreciation is deducted (subtracted) before taxes

# Depreciation Tax Shield

- Depreciation can also
  - act as a "shield" against and reduce corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)

- **Depreciation tax shield**, the tax saving that depreciation creates, as depreciation is deducted (subtracted) before taxes

$$\textbf{Depreciation tax shield} = \textbf{Tax rate} \times \textbf{Depreciation}$$

# Depreciation Tax Shield

- Depreciation can also
  - act as a "shield" against and reduce corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)
- **Depreciation tax shield**, the tax saving that depreciation creates, as depreciation is deducted (subtracted) before taxes

$$\textbf{Depreciation tax shield} = \textbf{Tax rate} \times \textbf{Depreciation}$$

- In the example, $21\% \times \$1{,}000 = \$210$

# Depreciation Tax Shield

- Depreciation can also
  - act as a "shield" against and reduce corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)
- **Depreciation tax shield**, the tax saving that depreciation creates, as depreciation is deducted (subtracted) before taxes

$$\textbf{Depreciation tax shield} = \textbf{Tax rate} \times \textbf{Depreciation}$$

- In the example, $21\% \times \$1{,}000 = \$210$
- Can a corporation have as much depreciation as possible? No, constrained by cost of acquisition and accounting rules.

# Math for Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by $1

# Math for Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate $(21\%) \times \$1 = ¢21$

# Math for Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate (21%) $\times$ \$1 = ¢21

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } ¢79} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\downarrow \text{ by } ¢21}$$

# Math for Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate (21%) $\times$ \$1 = ¢21

$$\underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by \$1}} - \underbrace{\text{Taxes}}_{\downarrow \text{ by ¢21}}$$

$$\underbrace{\text{CF from operating activities}}_{\uparrow \text{ by ¢21}} = \underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} + \underbrace{\text{Depreciation}}_{\uparrow \text{ by \$1}} - \underbrace{\text{Change in NWC}}_{\text{No change}}$$

# Math for Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate (21%) $\times$ \$1 = ¢21

$$\underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by \$1}} - \underbrace{\text{Taxes}}_{\downarrow \text{ by ¢21}}$$

$$\underbrace{\text{CF from operating activities}}_{\uparrow \text{ by ¢21}} = \underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} + \underbrace{\text{Depreciation}}_{\uparrow \text{ by \$1}} - \underbrace{\text{Change in NWC}}_{\text{No change}}$$

- Equity investors get ¢21 more

# Math for Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate (21%) $\times$ \$1 = ¢21

$$\underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by \$1}} - \underbrace{\text{Taxes}}_{\downarrow \text{ by ¢21}}$$

$$\underbrace{\text{CF from operating activities}}_{\uparrow \text{ by ¢21}} = \underbrace{\text{Net income}}_{\downarrow \text{ by ¢79}} + \underbrace{\text{Depreciation}}_{\uparrow \text{ by \$1}} - \underbrace{\text{Change in NWC}}_{\text{No change}}$$

- Equity investors get ¢21 more
- Each dollar of depreciation raises the *total* cash flow to investors by ¢21

# Practice Problem

- A corporation pays a 21% tax rate. Suppose it had an additional $200 million in depreciation. By how much would its taxes fall? What about its cash flow?

# Practice Problem

- A corporation pays a 21% tax rate. Suppose it had an additional \$200 million in depreciation. By how much would its taxes fall? What about its cash flow?
- Taxes fall by $21\% \times \$200 \text{ million} = \$42 \text{ million}$

# Practice Problem

- A corporation pays a 21% tax rate. Suppose it had an additional $200 million in depreciation. By how much would its taxes fall? What about its cash flow?
- Taxes fall by 21% $\times$ $200 million = $42 million
- Cash flow rises by the same $42 million, the depreciation tax shield

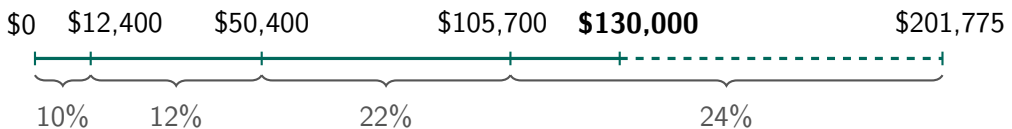
# Personal Taxation

- Personal tax rates matter for salaried employees and non-corporate businesses
- **Marginal tax rate**, the tax rate applied to each extra dollar of income
- **Average tax rate**, total taxes / total income

# Personal Taxation

| Taxable Income (Single Taxpayers, 2026) | Marginal Tax Rate (%) |
|---|---|
| \$0–\$12,400 | 10 |
| \$12,401–\$50,400 | 12 |
| \$50,401–\$105,700 | 22 |
| \$105,701–\$201,775 | 24 |
| \$201,776–\$256,225 | 32 |
| \$256,226–\$640,600 | 35 |
| \$640,601 and above | 37 |

Source, IRS Rev. Proc. 2025-32



# Example, Average Tax Rate

Suppose your salary is \$130,000 per year. What would be your average tax rate?

| Taxable Income (Single Taxpayers, 2026) | Marginal Tax Rate (%) |
|---|---|
| \$0–\$12,400 | 10 |
| \$12,401–\$50,400 | 12 |
| \$50,401–\$105,700 | 22 |
| \$105,701–\$201,775 | 24 |
| \$201,776–\$256,225 | 32 |
| \$256,226–\$640,600 | 35 |
| \$640,601 and above | 37 |

- Total tax $= \$12{,}400 \times 10\% + \$(50{,}400 - 12{,}400) \times 12\% + \$(105{,}700 - 50{,}400) \times 22\% + \$(130{,}000 - 105{,}700) \times 24\% = \$23{,}798$
- Average tax rate $= 23{,}798/130{,}000 = 18.31\%$

# Practice Problem

- Suppose your salary is \$30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

| Taxable Income (Single Taxpayers, 2026) | Marginal Tax Rate (%) |
|---|---|
| \$0–\$12,400 | 10 |
| \$12,401–\$50,400 | 12 |
| \$50,401–\$105,700 | 22 |
| \$105,701–\$201,775 | 24 |
| \$201,776–\$256,225 | 32 |
| \$256,226–\$640,600 | 35 |
| \$640,601 and above | 37 |

# Practice Problem

- Suppose your salary is \$30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

| Taxable Income (Single Taxpayers, 2026) | Marginal Tax Rate (%) |
|---|---|
| \$0–\$12,400 | 10 |
| \$12,401–\$50,400 | 12 |
| \$50,401–\$105,700 | 22 |
| \$105,701–\$201,775 | 24 |
| \$201,776–\$256,225 | 32 |
| \$256,226–\$640,600 | 35 |
| \$640,601 and above | 37 |

- Total tax $= \$12{,}400 \times 10\% + \$(30{,}000 - 12{,}400) \times 12\% = \$3{,}352$

# Practice Problem

- Suppose your salary is \$30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

| Taxable Income (Single Taxpayers, 2026) | Marginal Tax Rate (%) |
|---|---|
| \$0–\$12,400 | 10 |
| \$12,401–\$50,400 | 12 |
| \$50,401–\$105,700 | 22 |
| \$105,701–\$201,775 | 24 |
| \$201,776–\$256,225 | 32 |
| \$256,226–\$640,600 | 35 |
| \$640,601 and above | 37 |

- Total tax $= \$12{,}400 \times 10\% + \$(30{,}000 - 12{,}400) \times 12\% = \$3{,}352$
- Average tax rate $= 3{,}352/30{,}000 = 11.17\%$

# Practice Problem

- Suppose your salary is \$30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

| Taxable Income (Single Taxpayers, 2026) | Marginal Tax Rate (%) |
|---|---|
| \$0–\$12,400 | 10 |
| \$12,401–\$50,400 | 12 |
| \$50,401–\$105,700 | 22 |
| \$105,701–\$201,775 | 24 |
| \$201,776–\$256,225 | 32 |
| \$256,226–\$640,600 | 35 |
| \$640,601 and above | 37 |

- Total tax = $\$12{,}400 \times 10\% + \$(30{,}000 - 12{,}400) \times 12\% = \$3{,}352$
- Average tax rate = $3{,}352/30{,}000 = 11.17\%$
- Marginal tax rate = $12\%$

# Recap of Part 4

- The U.S. federal corporate tax rate is **21%**
- Interest and depreciation are deducted (subtracted) before taxes, so both shield income from tax
- A tax shield is worth the tax rate $\times$ the deduction, raising the *total* cash flow to investors
- Marginal rate applies to each extra dollar of income, average rate is total taxes over total income

# Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved

# Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved
- Book values are the cost of acquiring real assets or the amount raised from selling financial assets, market values are what things would sell for today

# Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved
- Book values are the cost of acquiring real assets or the amount raised from selling financial assets, market values are what things would sell for today
- Net income $\neq$ cash flow, add back depreciation, subtract the change in NWC

# Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved
- Book values are the cost of acquiring real assets or the amount raised from selling financial assets, market values are what things would sell for today
- Net income $\neq$ cash flow, add back depreciation, subtract the change in NWC
- Interest and depreciation shield taxes, raising the *total* cash flow to investors

# Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?

# Practice Problem

- A business pays a 21% tax rate. Its COGS is $20,000, its SG&A expenses are $10,000, and its depreciation is $8,000. Its interest expense is $2,000, and its taxes are $2,100. How much is its net income? How much is its EBIT? How much are its revenues?
- Taxable income = Taxes / Tax rate = $2,100 / 21% = $10,000

# Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?
- Taxable income = Taxes / Tax rate = \$2,100 / 21% = \$10,000
- Net income = Taxable income − Taxes = \$10,000 − \$2,100 = \$7,900

# Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?
- Taxable income = Taxes / Tax rate = \$2,100 / 21% = \$10,000
- Net income = Taxable income − Taxes = \$10,000 − \$2,100 = \$7,900
- EBIT = Taxable income + Interest expense = \$10,000 + \$2,000 = \$12,000

# Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?
- Taxable income = Taxes / Tax rate = \$2,100 / 21% = \$10,000
- Net income = Taxable income − Taxes = \$10,000 − \$2,100 = \$7,900
- EBIT = Taxable income + Interest expense = \$10,000 + \$2,000 = \$12,000
- Revenues = EBIT + COGS + SG&A expenses + Depreciation = \$12,000 + \$20,000 + \$10,000 + \$8,000 = \$50,000

# Practice Problem

If a firm's net income is positive and its depreciation is positive, which of the following could account for a negative cash flow from operating activities?

a. Inventories decrease while accounts receivable and accounts payable stay the same.
b. Accounts receivable increases while inventories and accounts payable stay the same.
c. The firm raises money by issuing stocks.
d. A large addition is made to plant and equipment.

# Practice Problem

If a firm's net income is positive and its depreciation is positive, which of the following could account for a negative cash flow from operating activities?

a. Inventories decrease while accounts receivable and accounts payable stay the same.
b. Accounts receivable increases while inventories and accounts payable stay the same.
c. The firm raises money by issuing stocks.
d. A large addition is made to plant and equipment.

# Thank you!

Please let me know if you have any questions.